

Two approaches to identifying fossil pollen

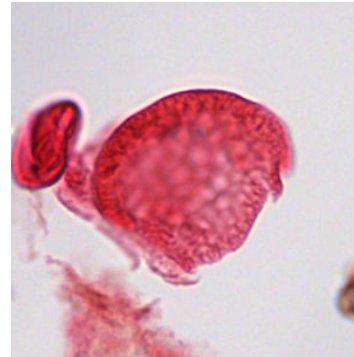
RF-DETR method first • classifier backbones second

01 RF-DETR

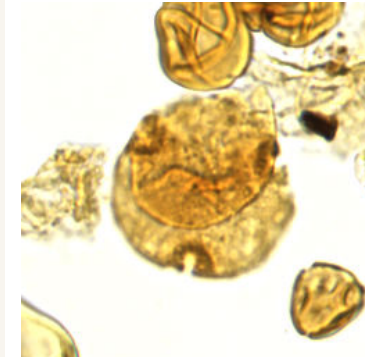
Find specimens and identify categories.

02 Classifier backbones

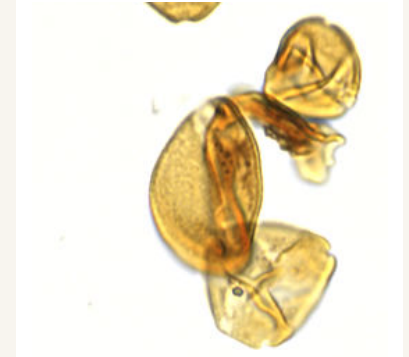
Identify categories from specimen crops.



Bisaccate



Bombacacidites sp.



Arecipites sp.

Classification dataset: 2,283 specimens • 32 priority categories

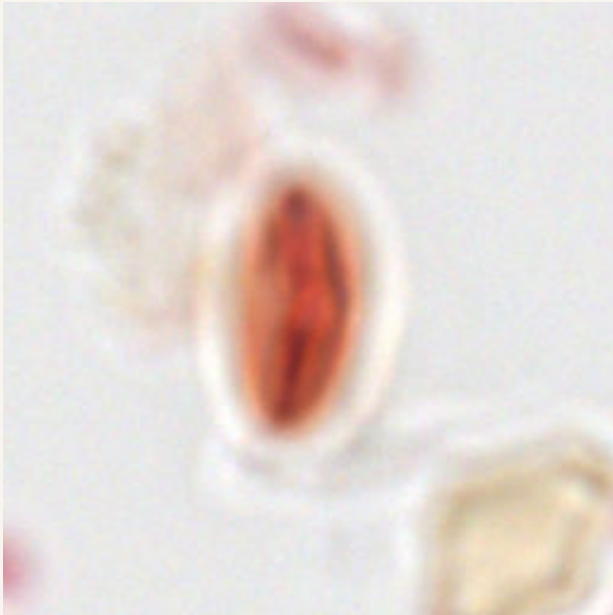
RF-DETR: detection and category identification

RF-DETR team update

Placeholder • content to be added

A clearer view helps the model identify pollen

Original focal plane



Selected sharper plane



Same specimen and crop • expert label: Quercus sp.

Unseen-slide accuracy

32% → 38%

Same Swin model settings.
Only the selected focal
plane changed.

Two questions, two ways to check performance

Unseen slides

Can it work on a different slide?

LEARN FROM

Slide A

Slide B

CHECK ON

Slide C

Slide D

Shared slides

Can it identify more grains on familiar slides?

LEARN FROM

A · grain 1

B · grain 1

CHECK ON

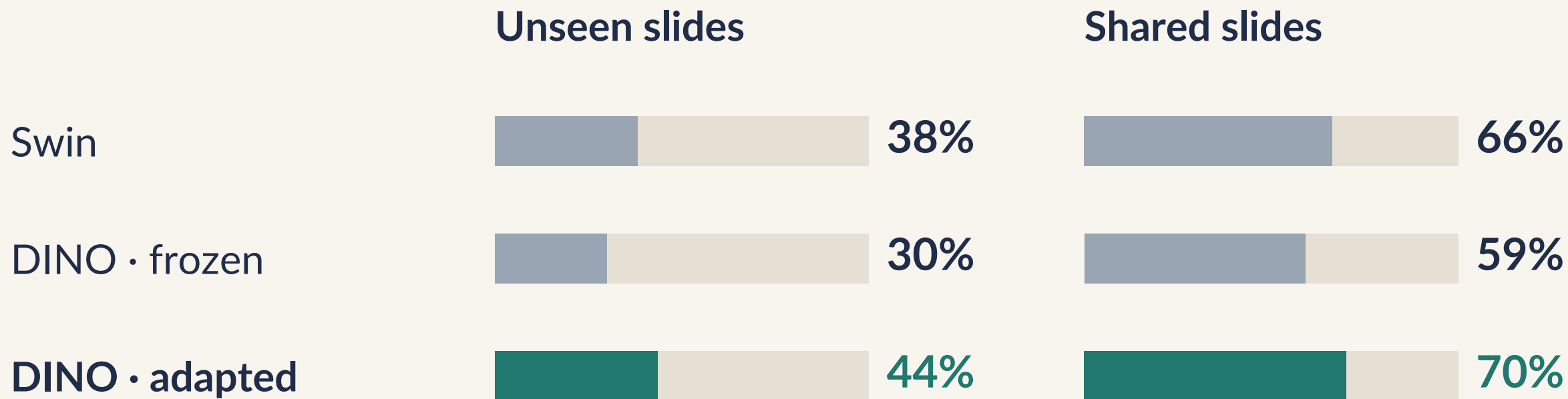
A · grain 2

B · grain 2

Different specimens in every set. Final test slides remain reserved.

Adapting DINO gives our strongest results so far

Validation accuracy · identifying categories in prepared specimen crops



Adapted = trained the final two DINO blocks and its category classifier.

Single-run validation comparisons; final test untouched. These are not whole-slide detection results.

Where identification still struggles

Adapted DINO · unseen-slide validation

EXPERT CATEGORY	CORRECTLY IDENTIFIED		CORRECT / TOTAL
Momipites wyomingensis			0% 0 / 21
Polyatriopollenites type			11% 1 / 9
Bombacacidites sp.			12% 2 / 17
Ulmipollenites sp.			16% 5 / 32
Psilatricolpites sp.			17% 3 / 18
Arecipites sp.			18% 2 / 11

18 of 21 Momipites wyomingensis were predicted as Momipites sp.

Lowest category recall among categories with ≥5 validation examples; small samples remain uncertain.